FAQ
Measuring Concentration and Purity

IONIC® Purification System

FAQs

Frequently Asked Questions Regarding Extract Concentration and Purity

VERSION 1.0 | DECEMBER 2020

ABOUT

This FAQ is intended to provide brief answers to questions regarding nucleic acid concentration and purity measurements of extracts from the Ionic® Purification System. For more information than what is provided here, please consult your Purigen Biosystems representative or email support@purigenbio.com.

Measuring the Concentration of Nucleic Acid in Ionic System Extracts

1 What method(s) for estimating DNA or RNA concentrations in Ionic extracts are recommended by Purigen Biosystems?

Analyte-specific (or intercalating) fluorescent dye (e.g. Qubit assays, ThermoFisher), qPCR, RT-qPCR, and ddPCR assays will accurately measure your nucleic acid concentration, or your amplifiable template concentration.

2 Can I use A260-based estimation from a microvolume spectrophotometer (e.g. ThermoFisher NanoDrop) of DNA or RNA concentrations in my Ionic System extracts?

Absorbance measurements can be more variable at lower concentrations of nucleic acid. Qubit High Sensitivity assays (ThermoFisher) are a better choice for quantification, using the properly matched analyte-specific kits. Exonuclease probe-based quantitative PCR is another alternative if a direct measurement of amplifiable material is relevant to downstream applications.

3 Why are microvolume spectrophotometers and Qubit (ThermoFisher) RNA and DNA concentrations for my Ionic System extracts different?

Microvolume spectrophotometers use absorbance-based assays that quantify all molecules that absorb at the set wavelength. The A260 measurement detects nucleic acids, protein, nucleotides, and free nucleosides that all absorb at 260 nm. Qubit assays, however, are fluorescence-based and use analyte-specific binding of dyes with characteristic shifts in emission spectrum between bound and unbound molecules (Stoke's shift). These dyes will not bind to nucleotides or other molecules that absorb light at 260 nm.

If the concentration of a sample measured by a microvolume spectrophotometer is higher than when measured by a Qubit assay, it is likely that additional substances present in your extract also absorb at 260 nm and are inflating the measurement.

Other studies have reported finding lower Qubit concentrations than microvolume spectrophotometer concentrations: https://www.nature.com/articles/s41598-017-08134-3.

4 Are NanoQuant Plates (Tecan) appropriate for concentration estimates of Ionic System extracts?

NanoQuant Plates can be used with either an absorbance-based assay or a fluorescence-based assay. Use of a fluorescence-based assay to measure the concentration of Ionic system extracts is recommended.

The NanoQuant can increase the throughput of measuring the concentration of nucleic acids that have been labeled with analyte-specific intercalating fluorescent dyes (e.g. Quant-iT™, PicoGreen™ dsDNA reagent, ThermoFisher). For this purpose, NanoQuant plates are recommended. Please note that using a fluorescent based assay can minimize the effect of sample contamination because of assay analyte specificity.

5 Can I use the BioAnalyzer (2100/2200, Agilent Technologies) to estimate the concentration of nucleic acid in my Ionic System extracts? Why are the nucleic acid concentrations measured by BioAnalyzer and Qubit different for my Ionic System extracts?

Using the BioAnalyzer 2100/2200 Lab Chips to estimate nucleic acid concentration is not recommended. Studies have found that while BioAnalyzer data gives valuable info on the DNA or RNA length profile, it fails to give reliable quantification of the concentration in solution:

- https://www.nature.com/articles/s41598-017-08134-3#ref-CR27
- https://www.nature.com/articles/s41598-017-08134-3
- https://www.future-science.com/doi/10.2144/000114403
- http://www.rrp.infim.ro/2016_68_3/A25.pdf

Considering the technical differences in these approaches, it is difficult to directly compare the BioAnalyzer results to other methods. BioAnalyzer concentration traces can be skewed by slight poor priming, improper ladder dilution and storage, pipetting, overloading, and dilution factors. In general, this assay is more sensitive to error. Qubit assays (intercalating dyes) on the other hand are generally less sensitive to such errors. An advantage of intercalating dye-based assays is that they are not affected by the typical components of complex biological mixtures and extraction reagents.

Measuring the Purity of Nucleic Acid in Ionic System Extracts

1 Does the low A260/A280 (e.g. NanoDrop) ratio of my Ionic System extracts mean that there is protein contamination in the extract?

A very low A260/280 ratio may indicate that the Ionic extracts have small peptide fragments that have carried over from the lysis reaction into the extract. Purigen has identified the Proteinase K stock solution used during the lysis procedure of Ionic System protocols is a source of peptide fragments using direct protein detection.

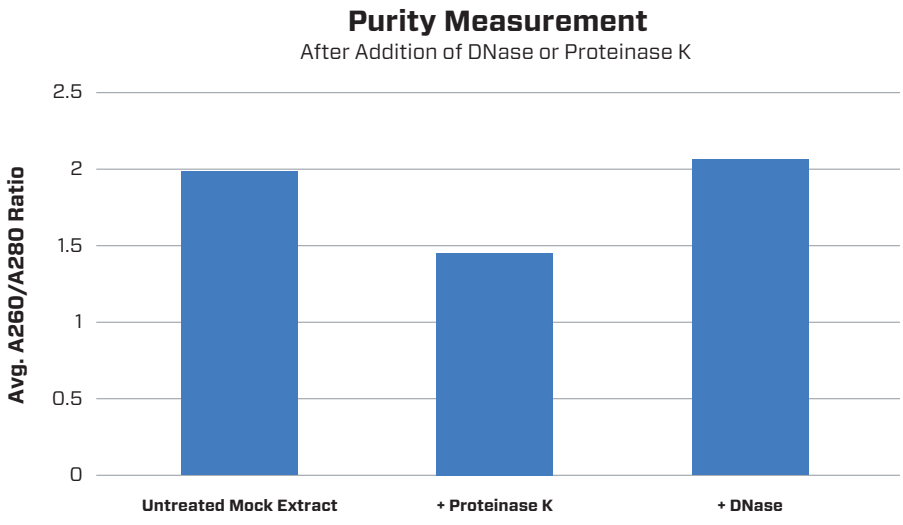


FIGURE 1: Assessment of purity after addition of lysis reaction components. DNase and Proteinase K was added to Ionic system extraction buffer containing high-quality RNA at concentrations expected after purification. The addition of Proteinase K significantly lowers the A260/280 ratio indicating Proteinase K as the source of peptide fragments in Ionic system extracts.

PURIGEN® BIOSYSTEMS

2 Will the low A260/A280 ratio of my Ionic System extract caused by peptide fragments interfere with NGS library preparation or sequence quality?

No, the peptide fragments causing the lower 260/280 ratio in Ionic extracts are inactive/degraded, and do not interfere with NGS library prep.

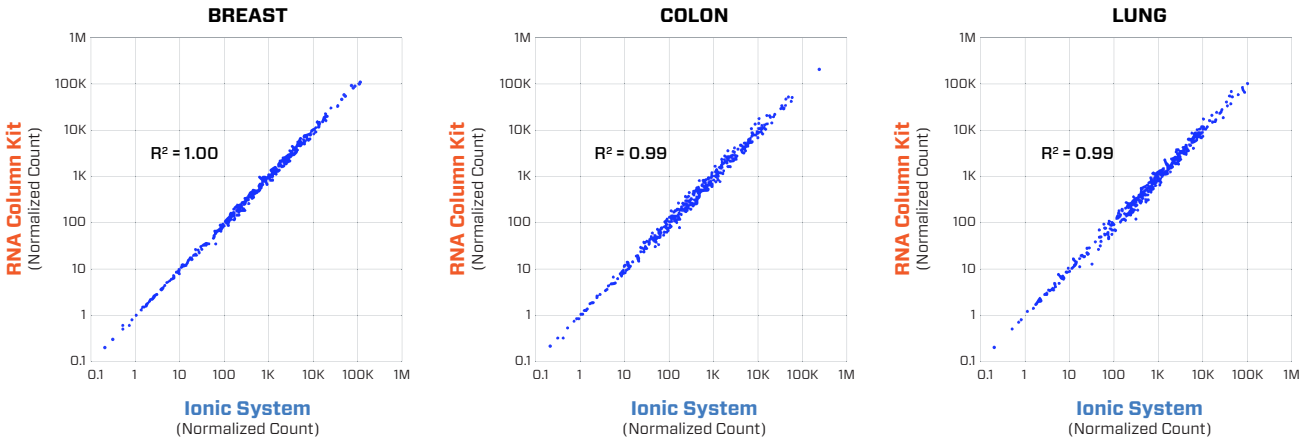


FIGURE 2: RNA sequencing results from samples purified by the Ionic system and a column-based kit. RNA was extracted and purified from replicate sections from three FFPE tissue blocks using the Ionic® FFPE to Pure RNA Purification Kit and a market-leading column-based kit for RNA extraction. NGS libraries were prepared for each extract following the protocol for the AmpliSeq for Illumina Immune Response Panel assay and sequenced on the Illumina MiSeq system. The Pearson correlation coefficient measure for each comparison indicates similar sequencing performance for each method.

3 What evidence does Purigen have that there is no residual protease activity or that the remaining protein fragments do not affect downstream assays such as qPCR, RT-qPCR or NGS library construction?

Purigen has investigated residual enzyme activity in Ionic extracts. The results demonstrate the absence of any active or intact residual protease in Ionic extracts (**FIGURE 3**, below). The Ionic system extracts from the enzyme-containing lysates, as well as enzyme-free lysates, produced fluorescence equivalent to the assay background levels.

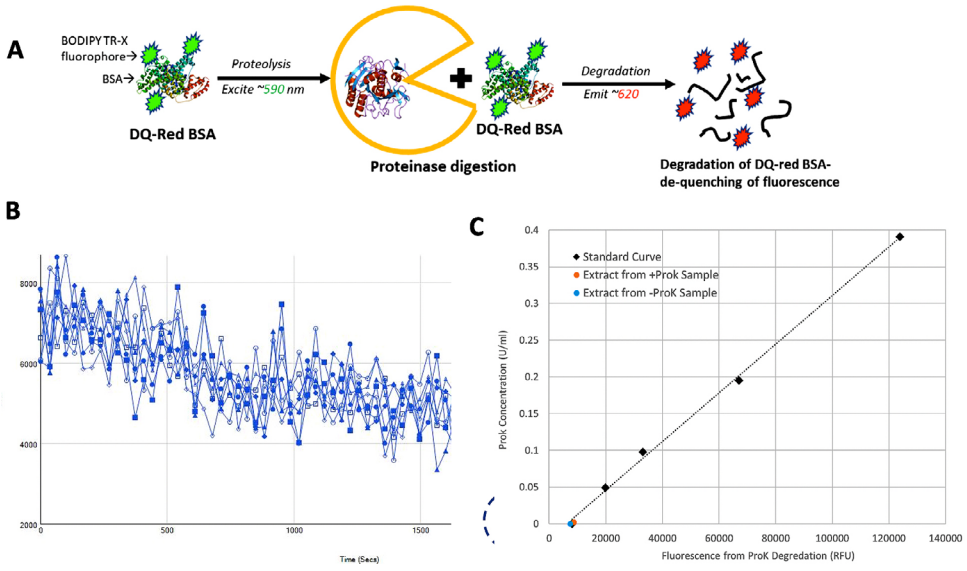


FIGURE 3: Testing of Ionic System extracts for residual protease activity. Ionic system extracts produced with and without enzymes expected to be present after lysis were tested for protease activity using the DQ-Red BSA reagent (ThermoFisher) (A). In this assay, the proteolysis of the BSA conjugates can be monitored due to the digestion resulting in dequenching by releasing fragments containing the only the fluorophores emitting at 620nm. 620nm intensity is measured over time. No difference was detected between real and enzyme-free lysate-derived extracts (B) or between extracts and negative controls (C).

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